

Geophysical Methods

Electrical
Resistivity
Tomography

Ground
Penetrating
Radar (GPR)

Magnetics

EM (Frequency Domain,
Time Domain)

Geophysical Methods

Method	What it measures	Main applications
ERT (Electrical Resistivity Tomography)	Electrical resistivity of the subsurface	Groundwater, contamination, shallow stratigraphy, foundation studies
EM (Electromagnetics)	Electrical conductivity (or resistivity) via induced currents	Conductive ore bodies, faults, groundwater, soil mapping
GPR (Ground Penetrating Radar)	Reflections of radar pulses at dielectric contrasts	Near-surface structures, buried objects, ice thickness, utilities
Magnetics (Ground-based)	Variations in the Earth's total magnetic field	Magnetic mineral mapping, igneous / metamorphic geology, faults

Geophysical Methods

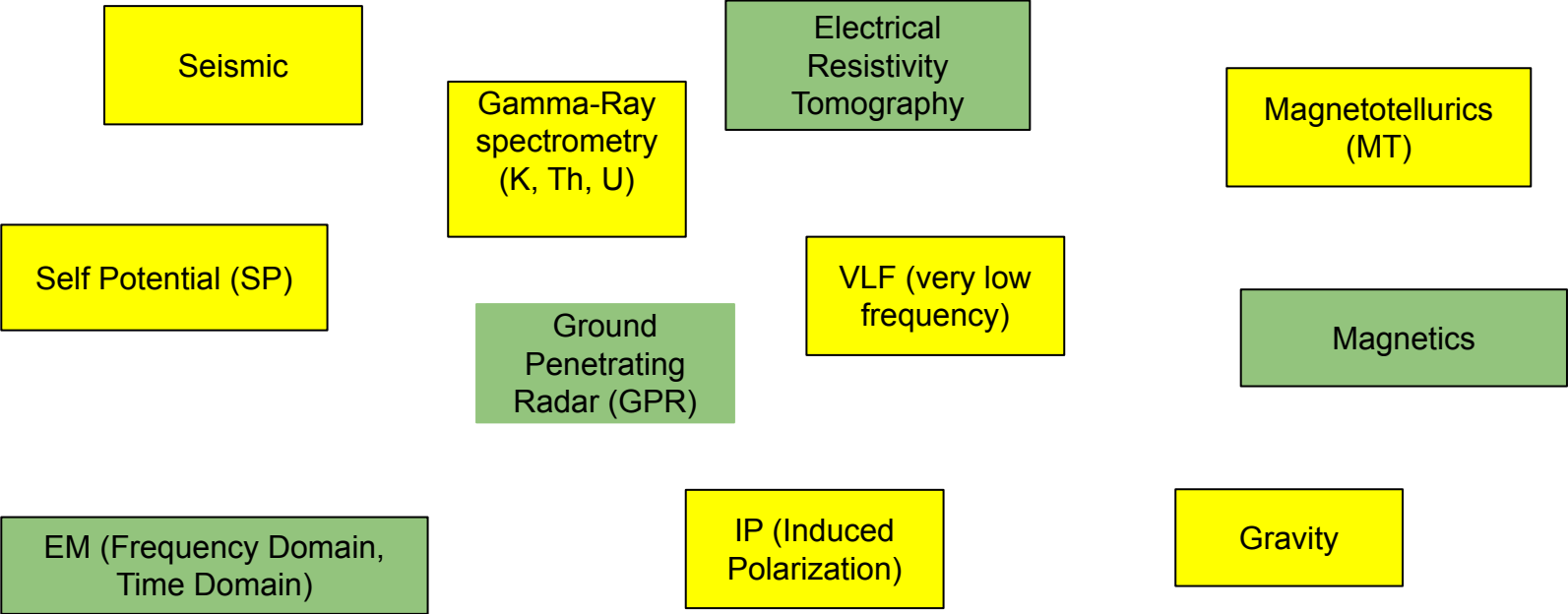
Electrical
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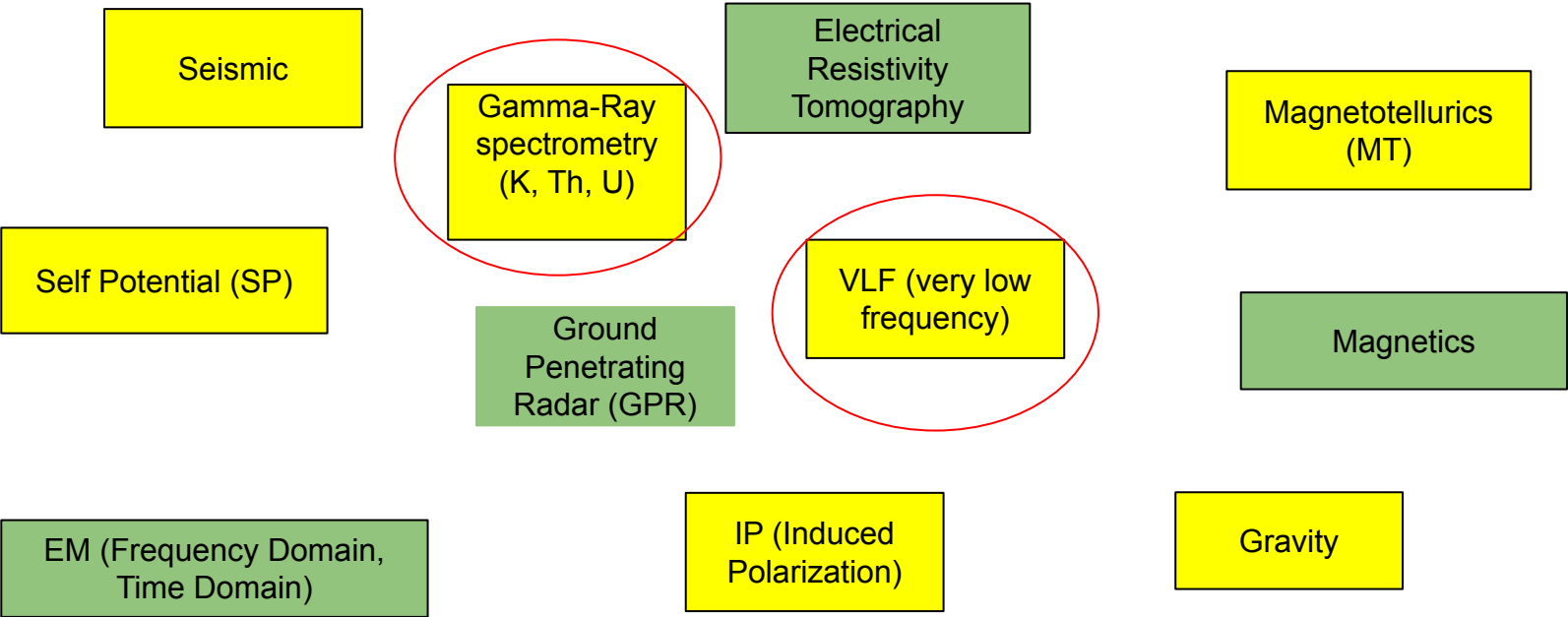
Magnetics

EM (Frequency Domain,
Time Domain)

Geophysical Methods



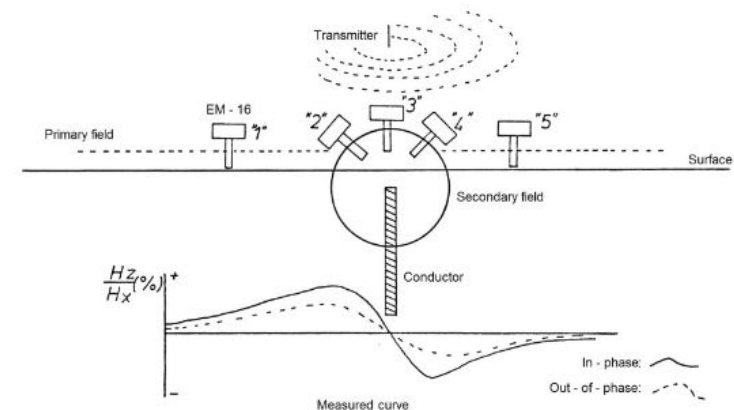
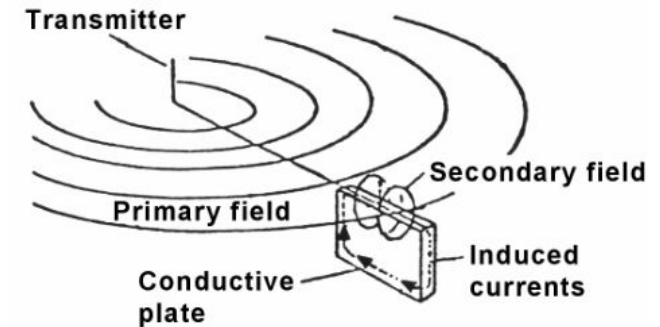
Geophysical Methods



VLF Method (Very Low Frequency)

Detection of conductive underground structures

- Detects conductive underground structures (e.g., fractures, mineral veins, groundwater, landfills).
- A VLF transmitter emits a low-frequency electromagnetic field (15–30 kHz).
- This field induces small currents in conductive bodies.
- The induced currents generate a measurable secondary field.
- Typical investigation depth: up to ~150 m. (depends on frequency and ground resistivity)
- Requires resistivity contrast between the target and surrounding materials.



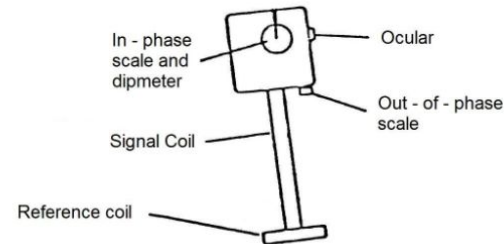
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VLF Method (Very Low Frequency)

An example of equipment used for ground-based surveys

- Handheld device (EM-16 by Geonics).
- Measures in-phase and out-of-phase components.
- Manual readings taken while walking along profiles.
- Portable method for small-scale surveys.

VLF Instrumentation (Geonics EM-16)



Ground water investigation with EM-16 (Geonics) in Jamaica

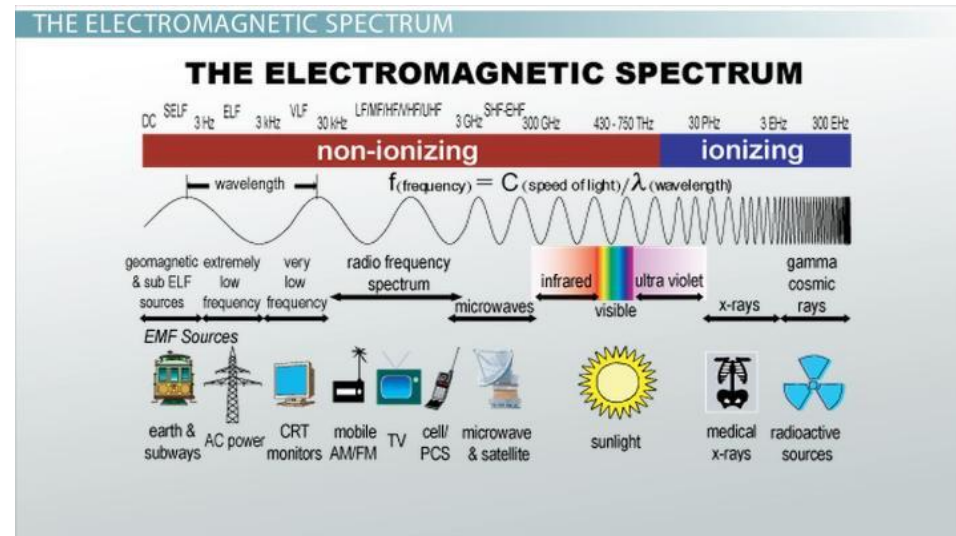
Gamma-Ray spectrometry (K, Th, U)

The Electromagnetic Spectrum: a range of energies

- The electromagnetic spectrum includes all types of electromagnetic waves, classified by their frequency and wavelength.
- The EM spectrum goes from radio waves (very low energy) all the way to gamma rays (highest energy)
- What separates them is frequency: the higher the frequency, the higher the energy

Different waves are generated by different physical processes:

- Radio waves: Electrons moving in antennas.
- Microwaves: Electrons oscillating at high frequencies.
- Infrared: Molecular vibrations (heat).
- Visible light: Electron transitions in atoms.
- Ultraviolet: High-energy electron transitions.
- X-rays: Sudden deceleration of electrons or deep electron transitions.
- Gamma rays: Nuclear decay processes.

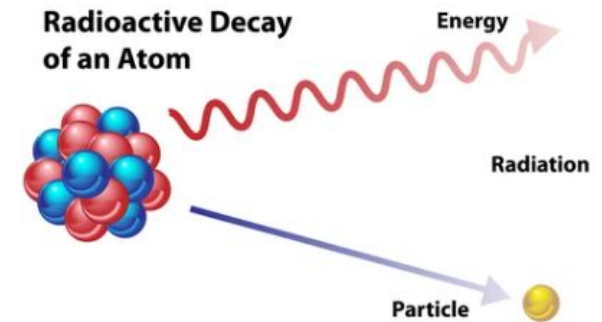


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Gamma-Ray spectrometry (K, Th, U)

What are Gamma-Rays

- Gamma rays sit at the far end of the spectrum: highest frequency, highest energy
- Unlike all other EM waves, they are not produced by electrons: they come from inside the nucleus
- They are released when an unstable nucleus reorganizes to reach a lower, more stable energy state: this is radioactive decay
- Because of their energy, they can penetrate matter: which is also why they are useful for geophysical surveys: they escape from the soil and reach our detector in the air



Gamma-Ray spectrometry (K, Th, U)

Gamma-Rays Spectrometry: We can detect Natural Radioactivity

On Earth, gamma rays are mainly emitted by the natural decay of radioactive isotopes such as Potassium (^{40}K), Uranium (^{238}U), and Thorium (^{232}Th).

These elements are naturally present in the Earth's crust and continuously release gamma radiation that can be detected by gamma-ray spectrometry.

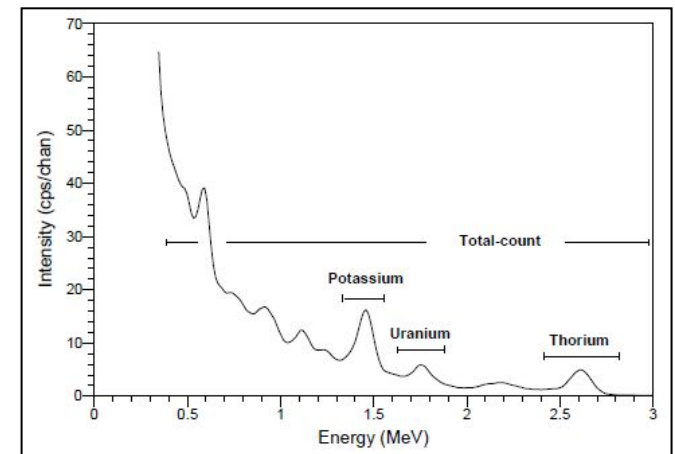
Each radioactive element emits gamma rays with characteristic energy levels:

- Potassium (^{40}K) $\rightarrow$ ~ 1.46 MeV
- Uranium (^{238}U) $\rightarrow$ ~ 1.76 MeV
- Thorium (^{232}Th) $\rightarrow$ ~ 2.62 MeV

These energy peaks are like unique fingerprints that allow us to detect and quantify the concentration of each element in the soil.

The results are typically reported as:

- Potassium (%)
- Equivalent Uranium (eU, ppm)
- Equivalent Thorium (eTh, ppm)



Gamma-Ray spectrometry (K, Th, U)

Gamma-Rays Spectrometry in geophysical surveys

The method is widely used for:

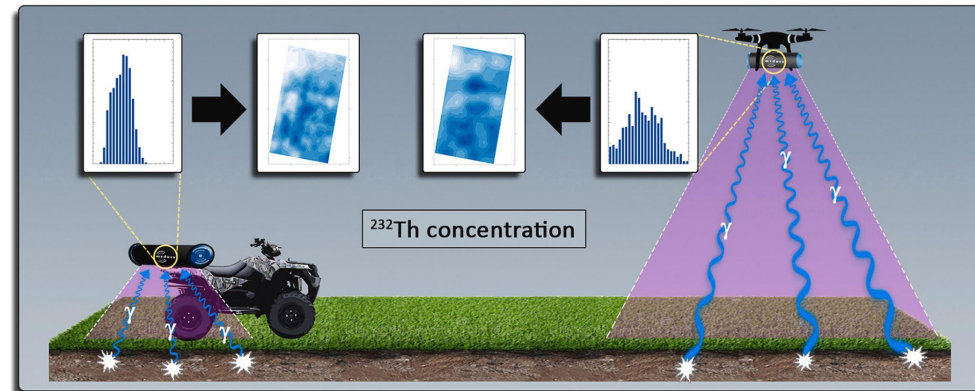
- Geological mapping to identify surface rock types and structures.
- Mineral exploration to detect potassium-rich or radioactive deposits.
- Environmental studies to assess soil composition and possible contamination.

Gamma-ray spectrometry can be applied in:

- Ground surveys using portable or vehicle-mounted sensors.
- Airborne surveys using aircraft or drones.
- Borehole logging to investigate subsurface formations.

The typical investigation depth is limited to the upper 30–40 cm of soil, where most gamma radiation originates.

Gamma-ray spectrometry provides fast, non-invasive surface data, making it ideal for large-scale surveys.



Airborne prospection

Why Airborne Geophysical Prospection?

Fast coverage of large areas.

Airborne surveys can explore vast and remote regions much faster than ground-based methods.

Access to challenging terrains.

Effective over forests, mountains, swamps, and other inaccessible or dangerous areas

Multi-sensor platforms.

Aircraft can carry magnetometers, EM systems, and radiometers in the same flight, providing integrated data.

High-resolution, continuous data.

Offers dense and consistent spatial coverage for detailed geological interpretation.

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Ground vs Marine vs Airborne

	Ground	Marine	Airborne
Coverage speed	Low (2-5 km/day)	Medium (ship, ~10-15 km/h)	High (up to 250 km/h)
Terrain access	Limited by relief and vegetation	Water bodies only	Any terrain (forest, desert, swamp)
Spatial resolution	Very high (point data)	High (continuous profile)	High and homogeneous (regular grid)
Cost per km ² (large areas)	Very high	High	Low

Airborne prospection

Why Airborne Geophysical Prospection?

What kind of aircraft is used in airborne geophysics?

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Platform	Strengths	Limitations
Fixed-wing (light aircraft)	Long range, fast cruise speed, large survey areas	Higher minimum speed, longer turning radius
Helicopter	Very low, slow flight, excellent drape over mountainous terrain	Higher cost per flight hour
Drone / UAV	Low cost, access to small or difficult areas	Limited range and payload



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Challenges in Airborne Magnetic Surveys

Magnetic disturbances from the aircraft.

Magnetic measurements are affected by the aircraft's structure and onboard systems like engines, generators, navigation devices, and radio equipment.

Platform motion effects.

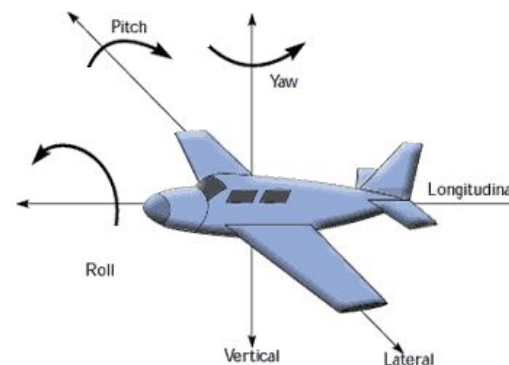
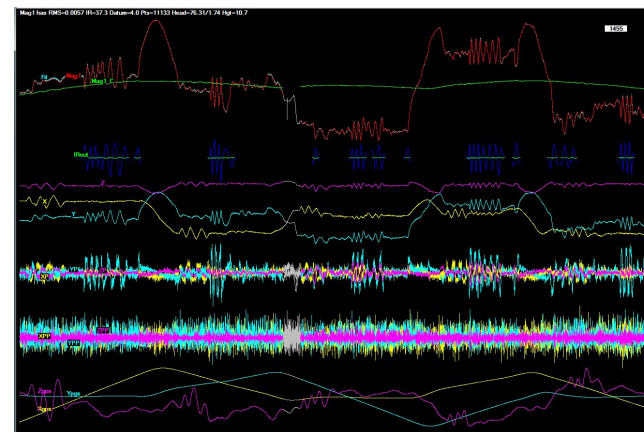
Aircraft maneuvers (pitch, roll, yaw) generate magnetic noise that must be corrected.

Sensor limitations.

Cesium vapor magnetometers have dead zones and heading errors that can cause data dropouts or sudden jumps.

Flight line design is essential.

Properly planned survey and tie lines enable later corrections such as tie line leveling.



Platform corrections - compensation

Compensation flight

Special maneuvers (compensation box or cloverleaf flight) are flown to measure and correct the aircraft's magnetic effects.

Lag correction.

Accounts for the distance and timing differences between the GPS and the sensor.

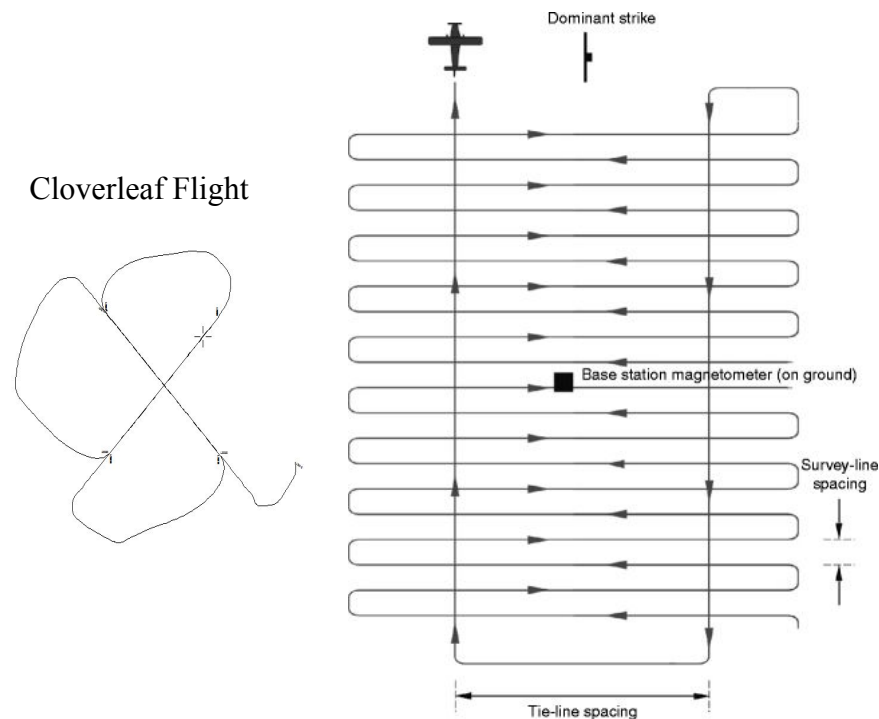
Heading correction.

Residual directional errors are corrected using a DC shift per flight direction.

Tie line leveling.

Adjusts data using control lines to remove systematic offsets.

Final goal: Obtain reliable magnetic data that truly reflects geological variations.

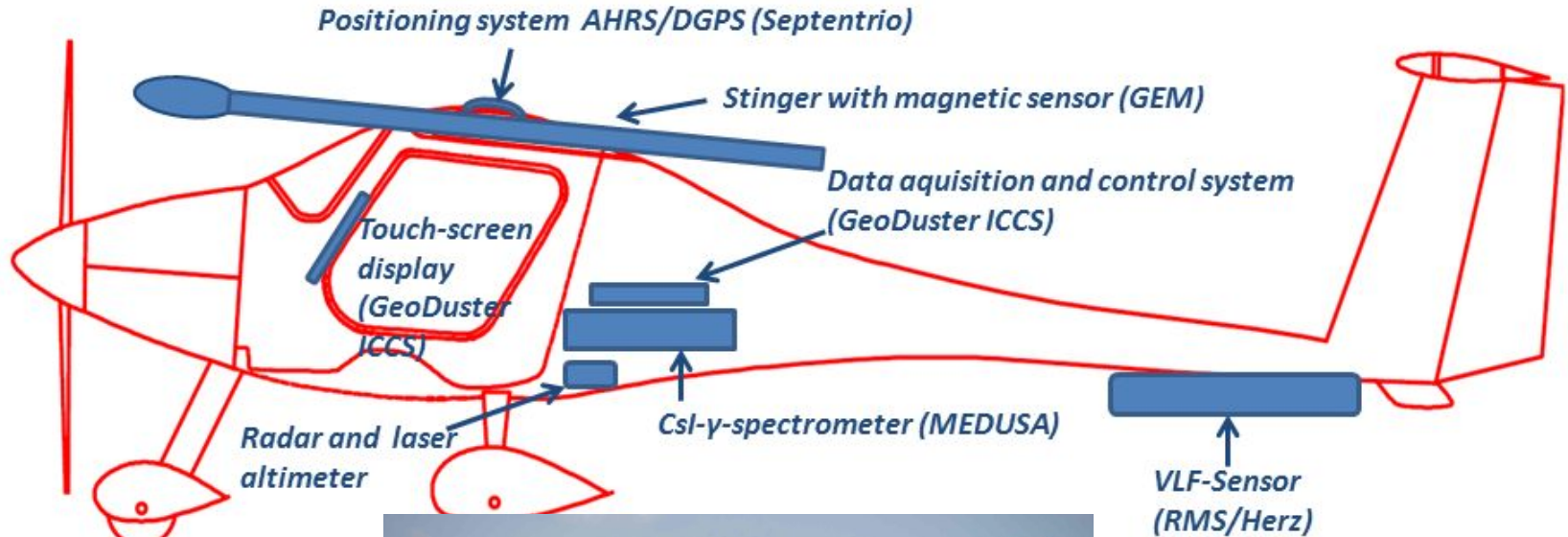


Airborne prospection

Onboard Equipment

System	Equipment	What it does
Positioning	AHRS / DGPS (Septentrio)	Position, altitude, attitude (roll / pitch / yaw)
Magnetometer	2x GEM K-GSMP 35	Mounted on a 'stinger', measures total magnetic field
Gamma spectrometer	Medusa, 4-litre Csl crystal	Radiometrics: K, U, Th (Full Spectrum Analysis)
VLF sensor	RMS / Herz	Detects conductive bodies in the subsurface

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Equipment:

- Compensation / Data acquisition System ICCS from GeoDuster
- 2 K-GSMP 35 airborne Magnetometers
- 1 γ -Spectrometer, with a 4 l CsI-Crystal
- 1 VLF EM receiver from DMS
- GPS
- Radiometer

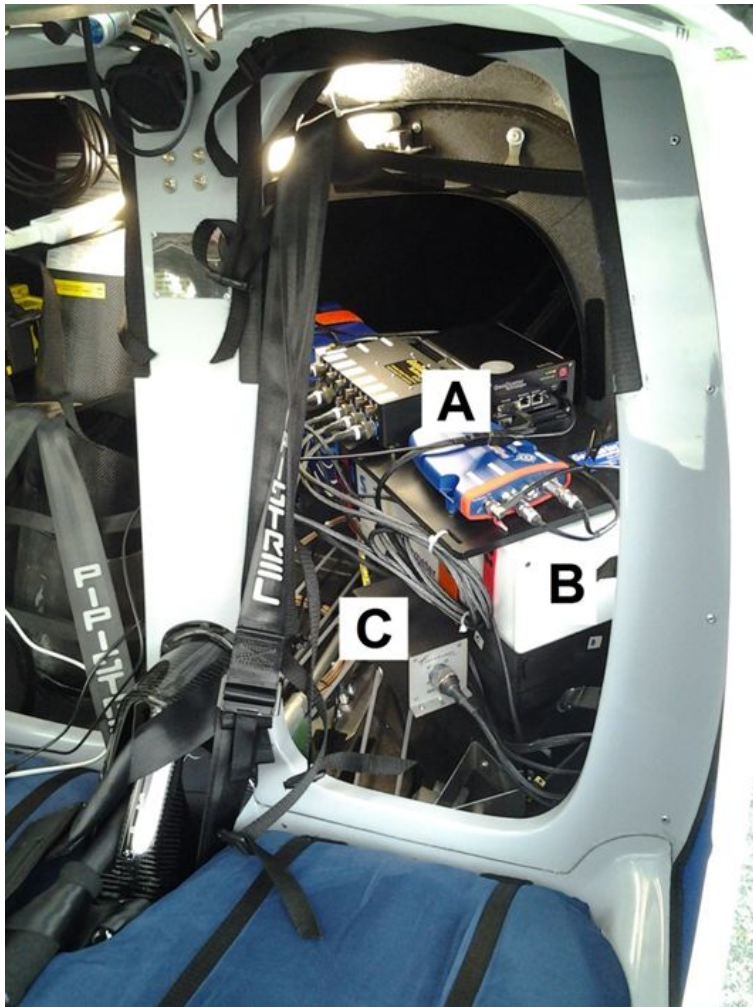


A white, high-wing, single-engine aircraft with a large propeller, parked on a grassy field. The aircraft has "NASA" and "VIRUS" logos on its side.

(specially designed as geophysical low “noise” platform)

- Full composite construction (carbon fibre)
 - Low weight, high stiffness, compact, easy demountable
- Low minimum speed ($V_{stall} = 65 \text{ km/h}$), good low speed flight characteristics
- High cruising speed ($V_{cruis.} = 250 \text{ km/h}$)
- Short take off run (95 m)
- Range 1700 km, Consumption 10 - 12 l/h MOGAS
- Cost effective (purchase, operation, maintenance)
- Aircraft rescue system

Insight:



Airborne geophysics in Mongolia - a new approach -



Conference GEOMINE
August 9-10, 2024 Ulaanbataar

Rainer Herd

Why airborne geophysics in Mongolia?

- Mongolia has a potential for different mineral resources
- But Mongolia is too big to be mapped **in detail** from the ground
- Traditional geological mapping can only find deposits which outcrop at the surface



Airborne geophysics (fast coverage of big areas and detection of mineralizations in the deeper underground)

What is the new approach?

Classic airborne geophysics:

- Standard metal aircraft $\longrightarrow$ relativ high magnetic noise
- Flying at fixed altitudes not taking into account the topography of the area $\longrightarrow$ different distances between sensors and ground
- Flying with high speed $\longrightarrow$ lower point density



not optimal data quality

New approach:

- Composite aircraft (carbon fibre) $\longrightarrow$ small magnetic noise
- Drap/contour flights $\longrightarrow$ always same distance between sensor and ground
- Flying with low speed $\longrightarrow$ high point density



improved data quality

The project

Background:

Agreement of Raw Material Partnership between Germany and Mongolia (signed 2011 by Angela Merkel and Sukhbaatar Batbold)

Aim:

- Test of the new airborne system under the conditions of Mongolia
- Detection of unknown mineral resources
- Cooperation of BTU, German-Mongolian Institute for Resources and Technology (GMIT) and the Mineral Resource and Petroleum Authority of Mongolia (MRPAM)
- Capacity building of Human Resources

Project diary

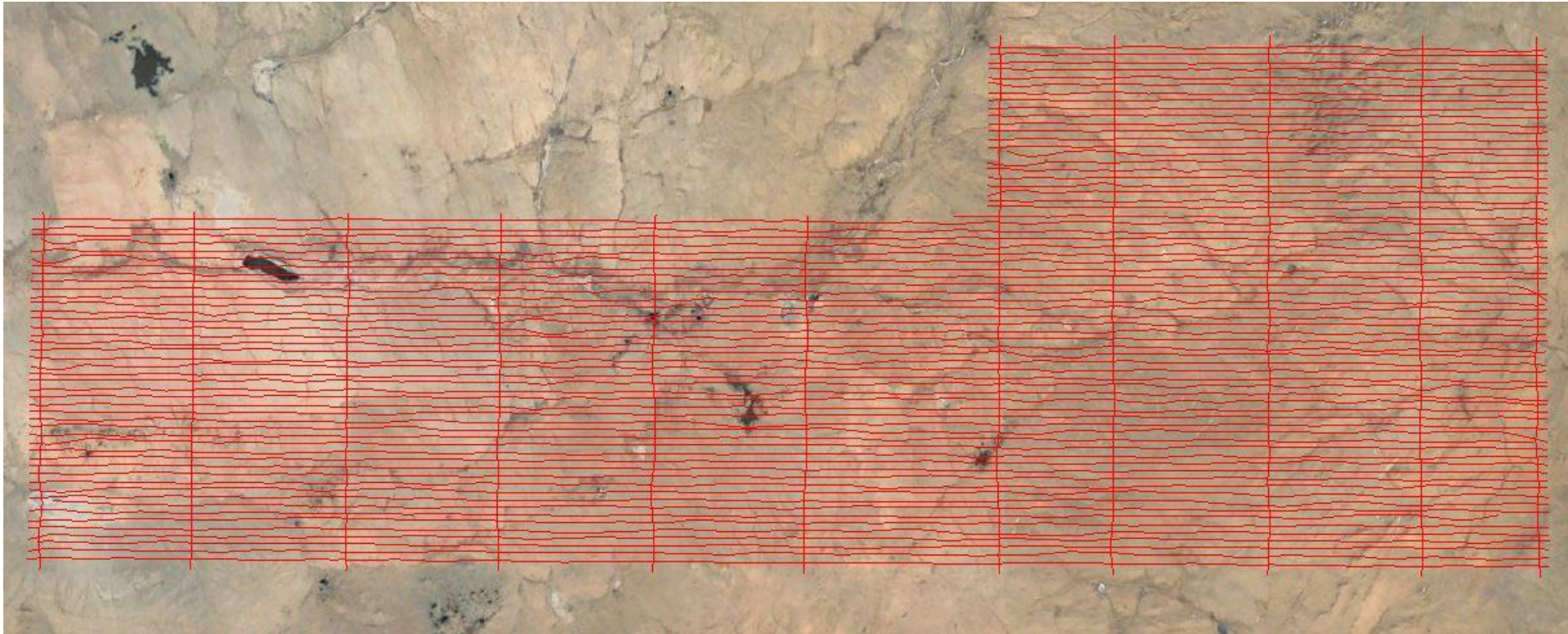


Project diary



12 09 2018

Example: Investigation area « Mandalgobi » (Central Gobi district)



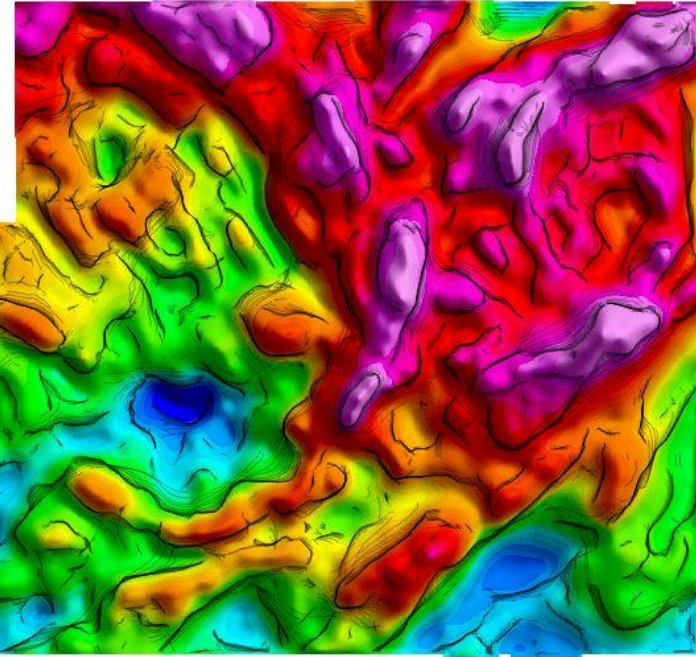
Areal extent: approx. 100 km²

Distance between lines: 100 m
Ground clearance: 100 m
Total: 1107 Line-km

[illegible]

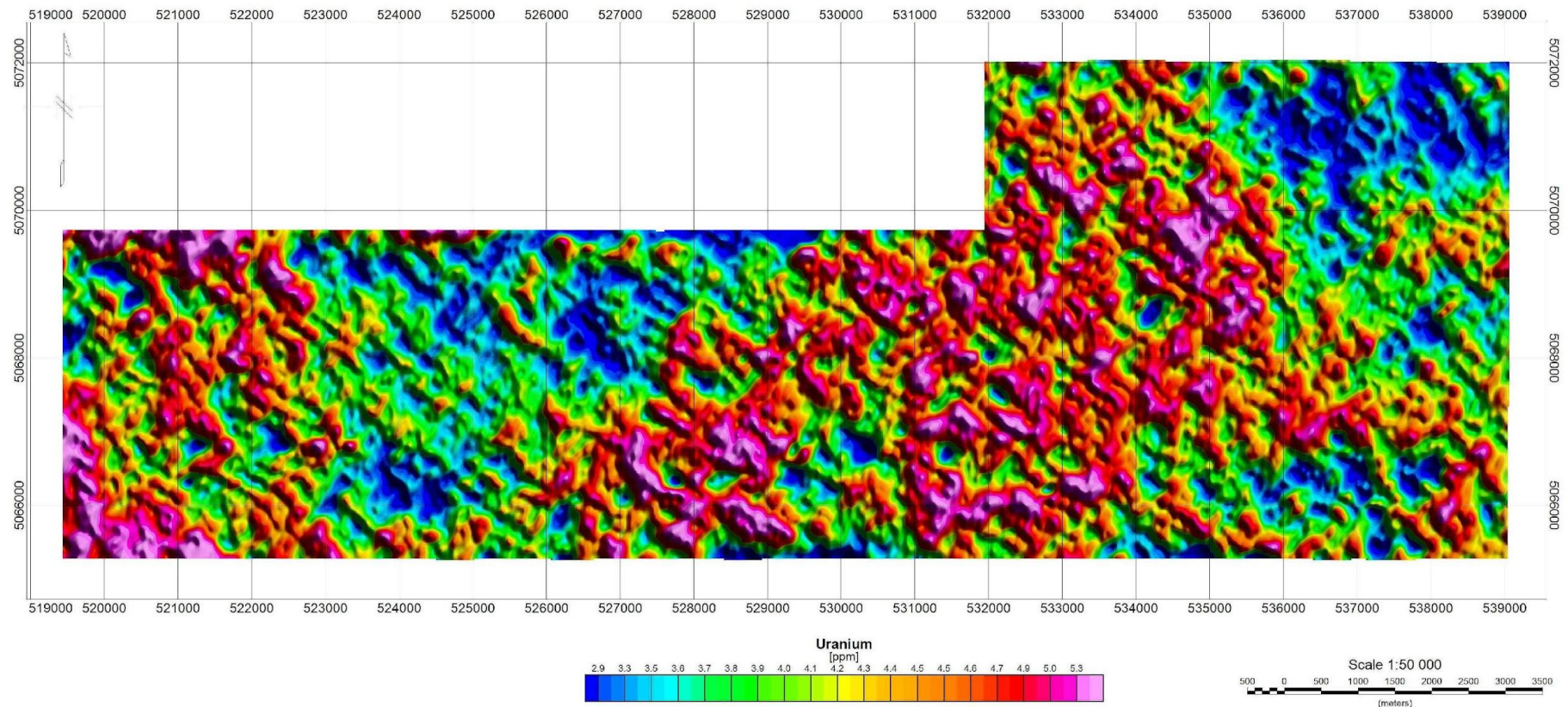
RTP
[nT]

37.3	93.0	135.6	189.6	244.1	280.5	311.7	338.5	364.9	394.4	435.6	486.5	544.1	678.5
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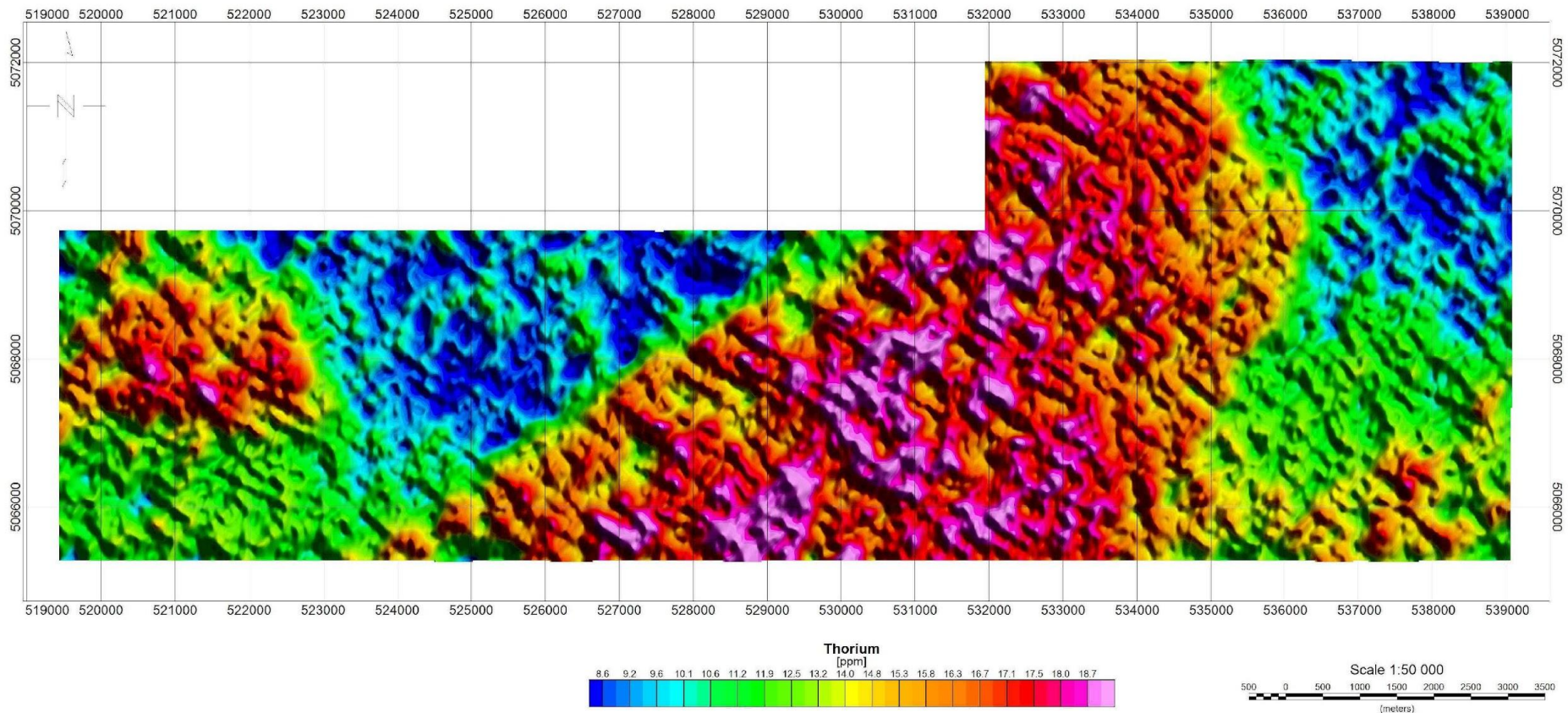
Investigation area « Mandalgobi » (Central Gobi district)

Results: Radiometrics (Uranium ppm)



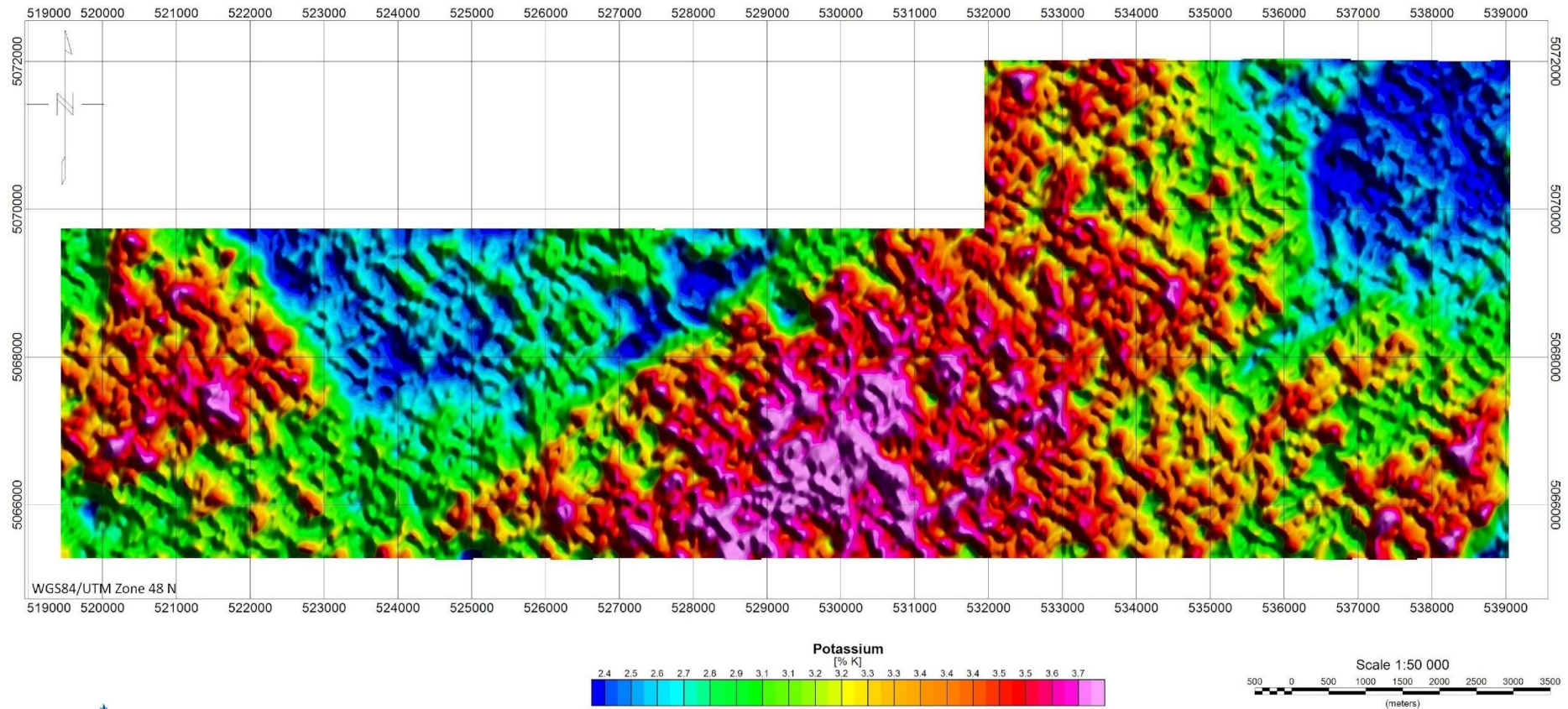
Investigation area « Mandalgobi » (Central Gobi district)

Results: Radiometrics (Thorium ppm)

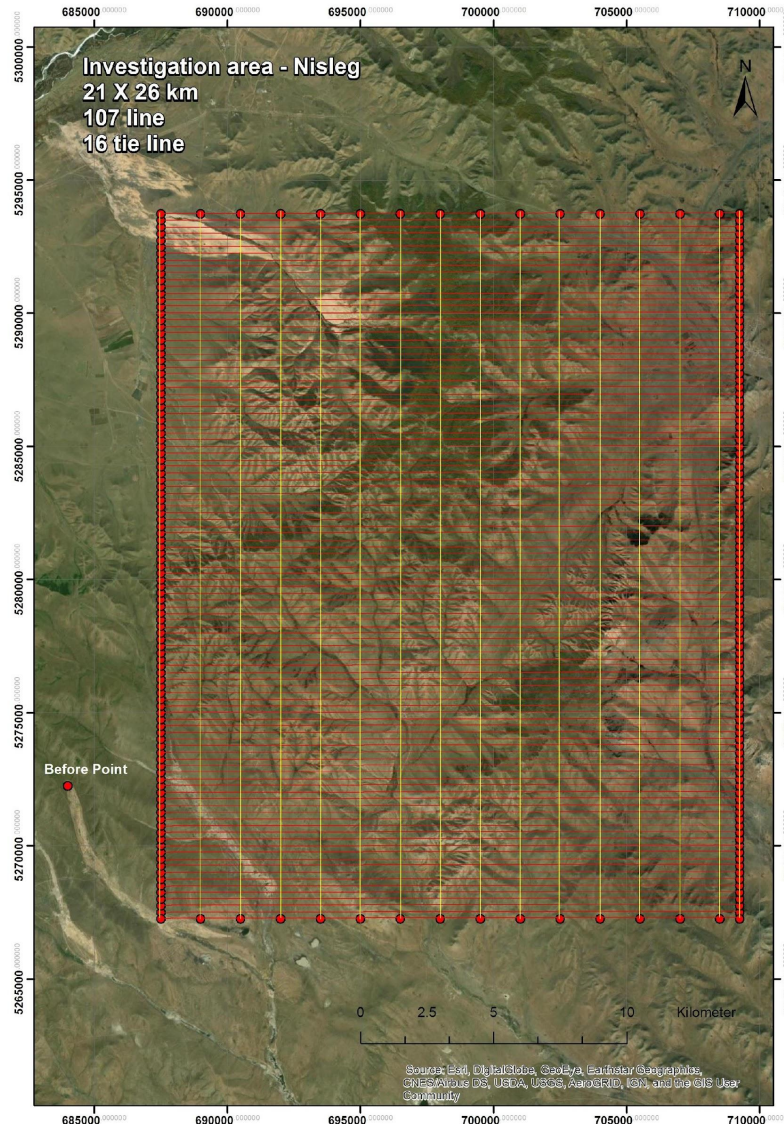


Investigation area « Mandalgobi » (Central Gobi district)

Results: Radiometrics (Potassium in %)



Investigation area « Janchiwlán »

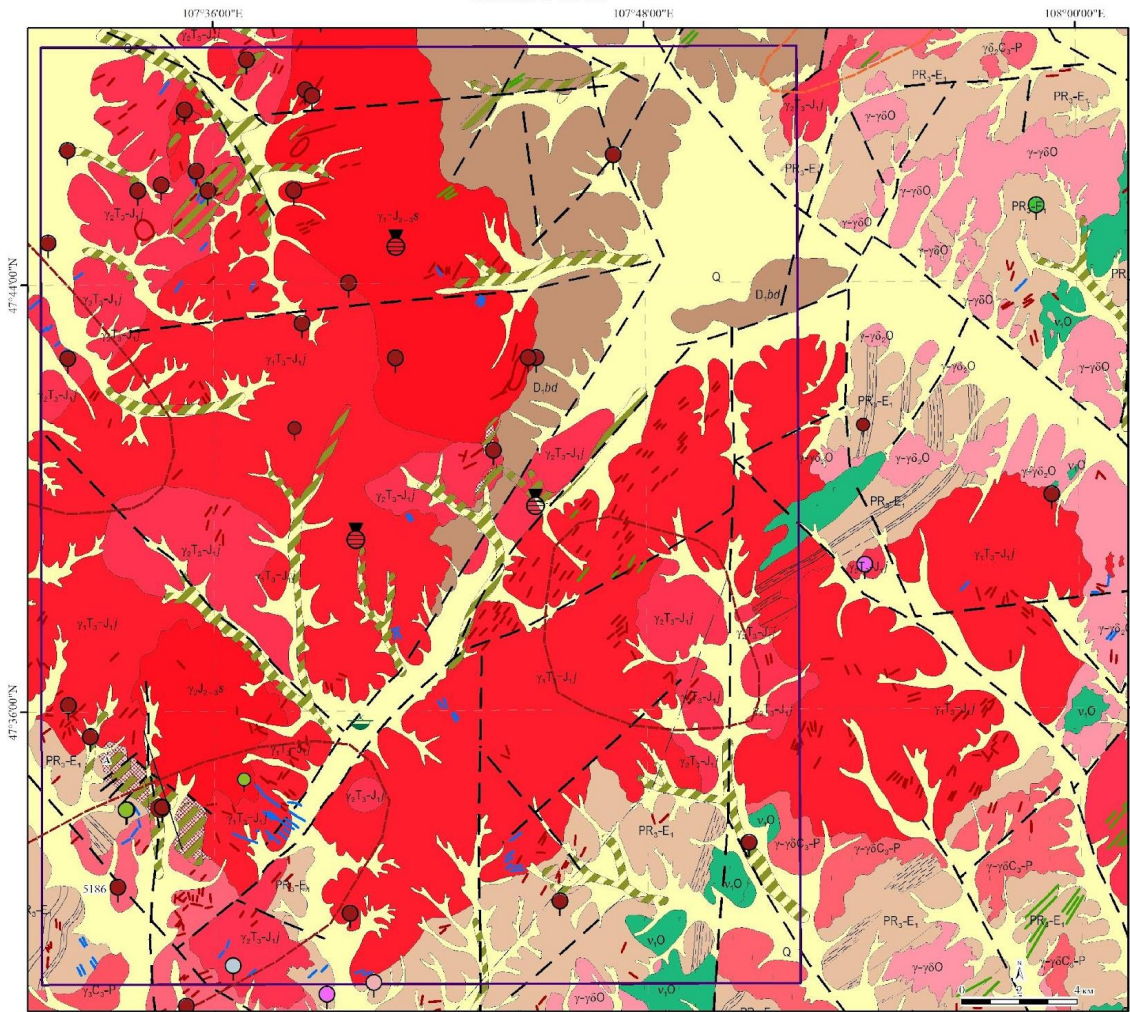


Altitudes from 1419 m a.m.s.l.
to 2041 m
Distance between lines: 250 m
Ground clearance: 100 m
Total: 2663 Line-km
Areal extent: 451 km²

Geology of area « Janchiwlan »

ЖАНЧИВЛАНГИЙН РОАЙНЫ ГЕОЛОГИЙН ЗУРАГ

Масштаб 1: 100 000



Тайлбар

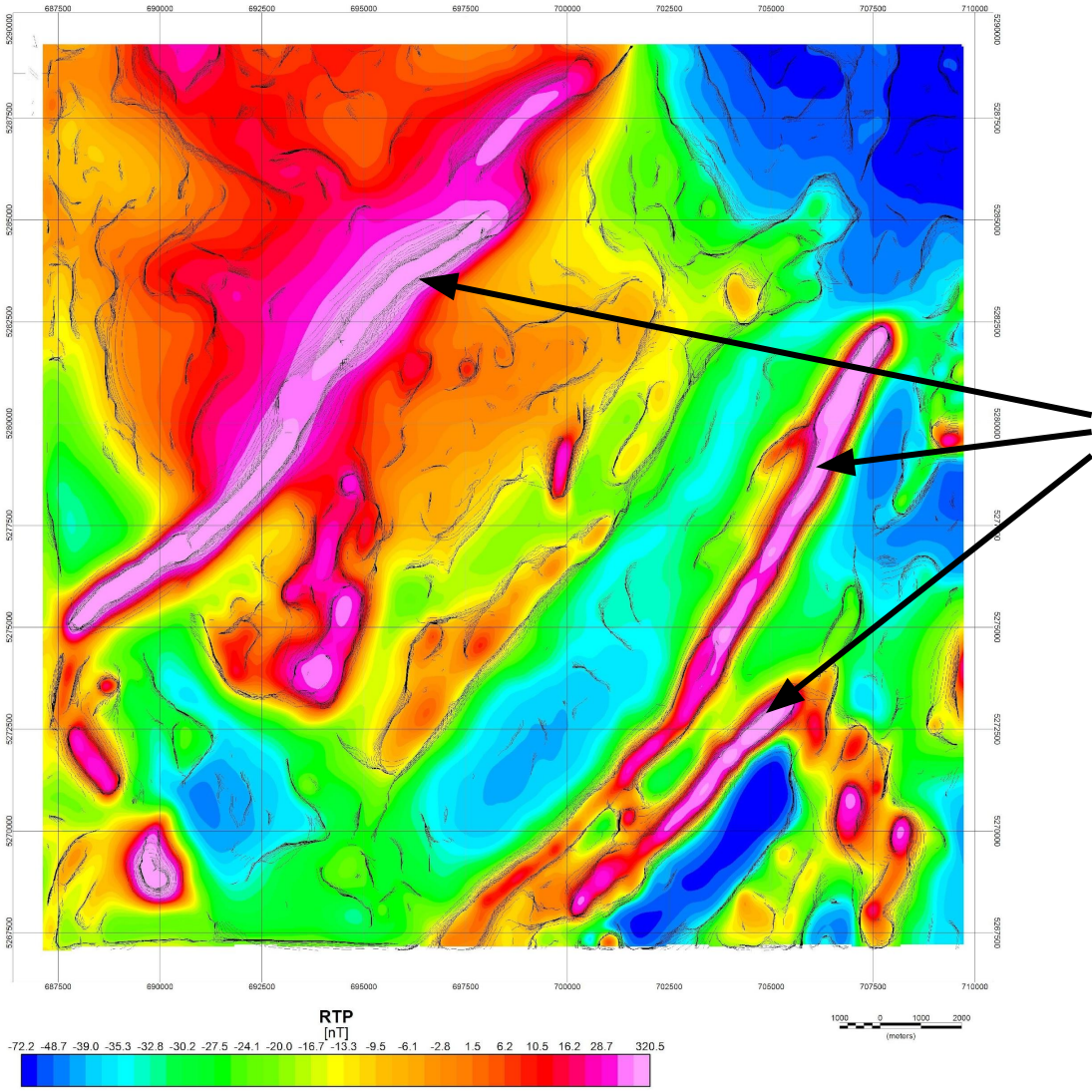
Данхрага зүй

- Q Дөрөвдөгч- орчин үеийн хурца
- D.bd Довол левон. Баяндаана форман, Ул суурийн конгломерат, гравелит, элсэн чулуу, алевролит, шохойн чулууны лит, ялтохид, пахурасат болог эффузив чулуулаг
- PR-6 Протерозой-доод кембри. Пахурасат, филлитийн занар, тунгала-тулганон чулуулууд, нолонит, арвант элсэн чулуу, тейс, амфиболит, шохойн чулууны лит нарийн үзүүртэй талст занар, хлорит-серпентиний, биотит-кварцтай занар

Интрузив чулуулаг

- γ-γδ, O Оровик. а. Биотиттой, дунд мөхлөгтэй катаклазжсан, хоорвоор асбкократ, гранодиорит, том мөхлөгтэй граносенит, хоорвоо адмакелит (γ-γδ₂) б. Дунд мөхлөгтэй габбро, габбро-диорит (γ₄)
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Results of Magnetics (area « Janchiwlan »)



In a well-known and mapped geological area unknown magnetic structures detected!

Impressions



Impressions



Outcome / Conclusion



- 3 areas with a total extend of 662 km² have been investigated, 4263 line-kilometers flown, **unknown structures in mapped areas could be detected**
- Ultra-light aircraft showed a **great robustness** and the investigation system a great performance
- Due to weather and climatic conditions the right survey time has to be chosen
- Airborne investigations with ultra-light aircrafts can be conducted **cost effective** in Mongolia

A photograph of three men standing on a paved airfield next to a white PIPISTREL VIRUS SV aircraft. The man on the left is wearing a tan flight suit. The man in the center is wearing a dark jacket and trousers. The man on the right is wearing a dark jacket and brown trousers. The aircraft has 'D-MTUC' on the side, along with 'VIRUS SV' and 'www.pipistrel.eu'. The background shows a dry, hilly landscape under a clear blue sky.

Ready for the next task

Thank you for your attention

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